

Simulating classical spin systems using the Corner Transfer Matrix Renormalization Group method.

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Abstract

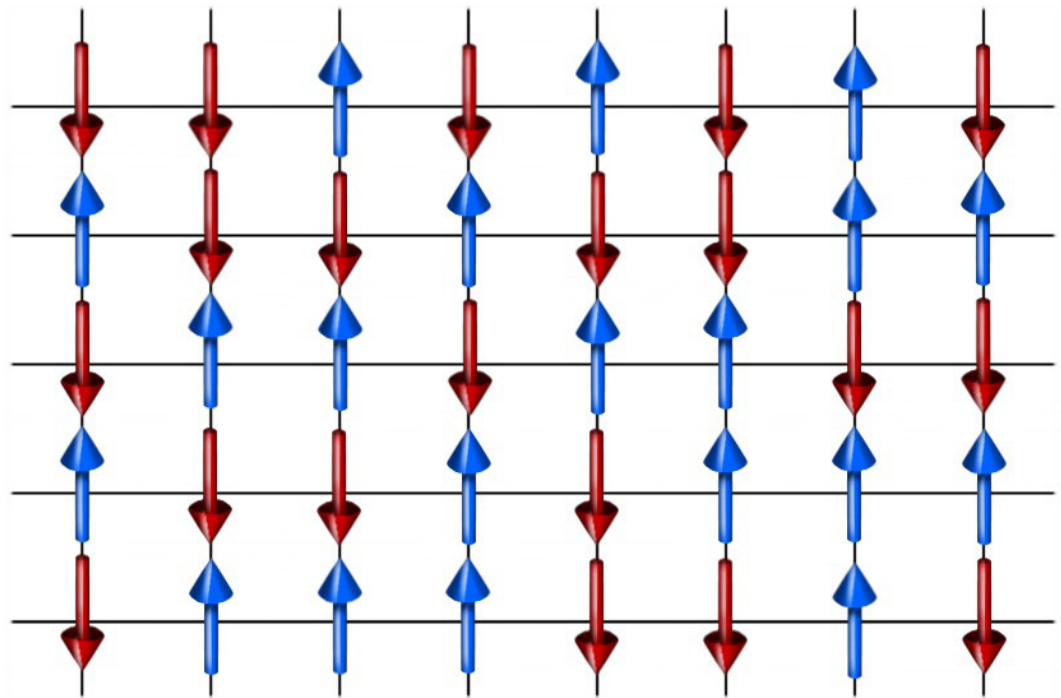
I use the Corner Transfer Matrix Renormalization Group method in order to simulate the partition function of the two-dimensional square lattice Ising and q -state Potts model. From the partition function I calculate the magnetization, free energy, specific heat and magnetic susceptibility per site of the Ising model. I find for the critical temperature of the Ising model that $T_c = 2.26920(1)$ and for the 3-state Potts model that $T_c = 0.99499(3)$. For the critical exponents of the specific heat, magnetization and magnetic susceptibility I determine the values $\alpha = 0$, $\beta = 0.1264(4)$ and $\gamma = 1.57(9)$ respectively.

Simulatie van de Ising model met behulp van tensoren.

In mijn onderzoek heb ik een model uit de statistische fysica onderzocht, genaamd het Ising model. Dit natuurkundige model beschrijft een twee-dimensionaal rooster met op elk rooster punt een deeltje dat in twee mogelijke toestanden kan zijn. In de afbeelding op de volgende pagina is zo'n configuratie weergegeven, waarbij de twee toestanden als een pijl omhoog en omlaag zijn afgebeeld. Elk deeltje wil het liefst in dezelfde toestand zijn als zijn dichtstbijzijnde buurdeeltjes, omdat de totale energie van het systeem dan minimaal is. Dit is alleen niet altijd mogelijk, omdat de deeltjes aangeslagen raken als de temperatuur ongelijk aan nul is en ze dan van toestand veranderen. Dit systeem is erg interessant, omdat er bij een bepaalde temperatuur een faseovergang plaatsvindt van een ongeordende fase, waarbij de toestanden van de deeltjes willekeurig zijn, naar een geordende fase, waarbij het merendeel van de deeltjes in dezelfde toestand zitten. Dit verschijnsel wordt 'spontane magnetisatie' genoemd en de temperatuur waarbij dit plaatsvindt wordt aangeduid met de kritische temperatuur.

De grootte van dit systeem is oneindig en daarom kost het oneindig veel tijd om het systeem op de computer te simuleren en exacte oplossingen te berekenen. Ik heb daarom een programma geschreven die het model benadert en bepaalde thermodynamische eigenschappen kan berekenen, zoals de magnetisatie, vrije energie, warmtecapaciteit en magnetische susceptibiliteit per deeltje. Ook heb ik het gedrag van deze grootheden bestudeerd vlakbij de kritische temperatuur. Mijn programma maakt gebruik van de 'corner transfer matrix renormalization group' methode. Hierbij wordt het systeem voorgesteld als een netwerk van tensoren. Tensoren zijn wiskundige objecten waarin getallen op een bepaalde manier opgeslagen zitten. De (oneindige) omgeving rond een bepaald rooster punt wordt vervolgens benaderd met omgevingstensoren door alleen de belangrijkste informatie van het systeem te bewaren. Deze tensoren kunnen vervolgens worden gebruikt om thermodynamische grootheden uit te rekenen.

Van veel thermodynamische grootheden zijn exacte oplossingen bekend. Deze heb ik gebruikt om de resultaten van mijn programma te controleren. Ik ben tot de conclusie gekomen dat mijn resultaten erg goed overeenkomen met de exacte oplossingen. De gebruikte methode is dus erg nauwkeurig en geschikt om de veralgemeniseerde versie van de Ising model te bestuderen, genaamd de Potts model. In dit model kunnen de deeltjes niet in twee, maar in q verschillende toestanden zitten. Voor dit model heb ik ook de magnetisatie onderzocht en ik heb gevonden dat de kritische temperaturen goed overeenkomen met de exacte oplossingen.



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1 Introduction

Investigation of the Potts model [1] has been of great interest since the mid twentieth century [2]. This model has a lot of interesting properties like its critical behavior and can be used to understand a variety of complex systems in lattice statistics. The q -state Potts model is a generalization of the 2-state Ising model [3] and was constructed shortly after Ashkin and Teller studied a 4-state version of this model [4].

In contrast to the two-dimensional (2D) Ising model, the 2D Potts model has no exact solutions, except for the critical temperature. Therefore this model has to be simulated on a computer. A big problem however is that the computation time scales exponentially with the system size. So calculating the exact partition function of the Potts model would take an infinite amount of time in the thermodynamic limit where the number of lattice sites go to infinity.

This problem can be solved by first representing the system by a tensor network. Kramer and Wannier used this in order to predict the critical temperature of the Ising model [5]. As an extension of their work Baxter formulated the Corner Transfer Matrix (CTM) method [6]. Based on Wilson's renormalization group method [7] White created the density matrix renormalization group (DMRG) method [8]. Only two decades ago, Nishino and Okunishi combined the methods of Baxter and White, and created the corner transfer matrix renormalization group (CTMRG) method [9], which is a fast numerical method that can be used to analyze the Potts model.

In this report I show how I use the CTMRG method in order to simulate the partition function of the 2D Ising model. With the partition function I calculate other thermodynamic quantities of the system, like the magnetization, free energy, specific heat and magnetic susceptibility per site. I also determine the critical temperature and the critical exponents of the specific heat, magnetization and magnetic susceptibility. After this I apply the method to the Potts model in order to study the magnetization.

2 Theory

2.1 The Ising model

2.1.1 Definition of the Ising model

The Ising model is invented by Wilhelm Lenz in 1920 and the 1D problem is five years later solved by his student Ernst Ising [3]. The 2D Ising model describes an infinite two-dimensional lattice with a spin on each lattice site which can either have the value +1 or -1. These values can be seen as state up and down respectively. The Hamiltonian of this system consists of two parts and is given by

$$H(\{s_i\}) = - \sum_{\langle i,j \rangle} J_{ij} s_i s_j - \sum_i h_i s_i. \quad (1)$$

The Hamiltonian is completely determined by the spin configuration $\{s_i\}$ on the lattice. The first part describes the interaction energy between the spins. The notation $\langle i, j \rangle$ means that the sum goes over all pairs of nearest neighbor spins. J_{ij} is a constant that specifies the strength of the interaction between spin i and spin j (s_i and s_j). The second part of the Hamiltonian is a sum over all spins that describes the interaction of each spin s_i with the external magnetic field h_i . If the interaction strength is the same between each spin and the magnetic field is uniform, the Hamiltonian can be written as

$$H(\{s_i\}) = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i. \quad (2)$$

The sign of J has a physical meaning. When $J > 0$ the neighboring spins prefer to be in the same state and then the system acts as a ferromagnet. When $J < 0$ the neighboring spins prefer to be in different states and then the system acts as an anti-ferromagnet. Obviously, when there is no external magnetic field, the second part of the Hamiltonian in equation 2 will disappear.

When working in the canonical ensemble, the partition function of the Ising model can be expressed as

$$Z(\beta) = \sum_{\{s_i\}} e^{-\beta H(\{s_i\})} = \sum_{\{s_i\}} e^{\beta J \sum_{\langle i,j \rangle} s_i s_j + \beta h \sum_i s_i} \quad (3)$$

$$= \sum_{\{s_i\}} \prod_{\langle i,j \rangle} e^{\beta J s_i s_j} \prod_i e^{\beta h s_i}, \quad (4)$$

where β is the inverse temperature $\frac{1}{k_B T}$, with k_B the Boltzmann's constant and T the temperature. From the partition function it is possible to determine all other thermodynamic quantities. The magnetization m , free energy f , specific heat c_v and magnetic susceptibility χ_m per site are given by [10]

$$m = \langle s_r \rangle = \frac{1}{Z} \sum_{\{s_i\}} s_r \prod_{\langle i,j \rangle} e^{\beta J s_i s_j} \prod_i e^{\beta h s_i}, \quad (5)$$

$$f = -k_B T \ln(Z)/N, \quad (6)$$

$$c_v = \frac{1}{N k_B T^2} \frac{\partial^2 \ln(Z)}{\partial \beta^2} \quad \text{and} \quad (7)$$

$$\chi_m = \left. \frac{\partial m}{\partial h} \right|_{h=0}, \quad (8)$$

where s_r is the spin of lattice site r and N is the number of lattice sites.

An interesting property of the 2D Ising model is that the system gets magnetized when the temperature goes below a certain critical temperature T_c , even when there is no external magnetic field. At this temperature the system undergoes a continuous phase transition from the disordered (non-magnetized) to ordered (magnetized) phase. Ordered means that a majority of the spins is in the same state (up or down). This phenomenon is called spontaneous magnetization. The Hamiltonian has a discrete symmetry: when all spins are flipped, the energy of the configuration does not change. Below T_c this symmetry is spontaneously broken.

Because the spins are interacting with each other, they are correlated in some way. The correlation between two spins s_i and s_j at different positions i and j is described by the correlation function, which has the form

$$C(i, j) = \langle s_i s_j \rangle - \langle s_i \rangle \langle s_j \rangle = \langle s_i s_j \rangle - \langle s_i \rangle^2 \quad (9)$$

$$= \langle s_i s_j \rangle - m^2. \quad (10)$$

For temperatures other than the critical temperature, the correlation function decays exponentially as the distance between s_i and s_j goes to infinity, according to

$$C(i, j) \sim \exp(-|i - j|/\xi), \quad (11)$$

where ξ is the correlation length. ξ is a measure of the size of the domains of aligned spins at a certain temperature. Above the critical temperature these domains are small because of thermal fluctuations. At the critical temperature there are domains with a variety of sizes, so there is no particular correlation length. When the temperature decreases below the critical temperature the domains grow until all spins are aligned at zero temperature.

2.1.2 Critical exponents of the 2D Ising model

In the Ising model, all thermodynamic quantities show critical behavior [11] near the critical temperature. Typical for a continuous phase transition, the slope of the magnetization goes to infinity when $T \rightarrow T_c$. The specific heat, magnetic susceptibility and correlation length diverge as well. It is known that these quantities behave near the critical temperature with a power law, each with its own critical exponent, according to

$$c_v \sim |t|^{-\alpha} \quad (12)$$

$$m \sim t^\beta \quad (13)$$

$$\chi_m \sim |t|^{-\gamma} \quad (14)$$

$$\xi \sim |t|^{-\nu}, \text{ where } t = (T - T_c)/T_c. \quad (15)$$

The β appearing must not be confused with the inverse temperature. These critical exponents are related to each other. Together they follow the scaling law

$$\alpha + 2\beta + \gamma = 2. \quad (16)$$

The values for α , β , γ and ν depend on the dimensions and symmetry of the system. For the 2D square lattice Ising model, they have the values [12]

$$\alpha = 0, \quad \beta = \frac{1}{8}, \quad \gamma = \frac{7}{4} \quad \text{and} \quad \nu = 1. \quad (17)$$

The specific heat does not follow a power law in this case, but diverges logarithmically;

$$c_v \sim \ln |t|. \quad (18)$$

2.1.3 Exact solutions of the 2D square lattice Ising model

Ernst Ising solved the 1D Ising model in 1925 [3], but the 2D Ising model was a lot more difficult. In 1944 Lars Onsager managed to give exact solutions for the magnetization and free energy per site of the 2D square Ising model in zero magnetic field ($h = 0$) [13]. He did not give a derivation of the exact solution for the spontaneous magnetization, but this was given three years later by Yang [14]. The exact solutions are given by

$$m = [1 - \sinh^{-4}(2\beta J)]^{\frac{1}{8}} \quad \text{and} \quad (19)$$

$$-\beta f = \frac{\ln(2)}{2} + \frac{1}{2\pi} \int_0^\pi \ln \left[\cosh^2(2\beta J) + \frac{1}{k} \sqrt{1 + k^2 - 2k \cos(2\theta)} \right] d\theta, \quad (20)$$

where $k = \sinh^{-2}(2\beta J)$. From equation 19 it is clear at which temperature the system undergoes a phase transition. By equating the magnetization per site to zero, you find that the critical temperature is:

$$T_c = \frac{1}{k_B \beta_c} = \frac{2J}{k_B \sinh^{-1}(1)} \approx 2.269 J/k_B \quad (21)$$

2.1.4 Tensor network representation of the 2D Ising model

For a 2D square lattice, the partition function in equation 4 can be written more compactly using that

$$\prod_i e^{\beta h s_i} = \prod_{\langle i,j \rangle} e^{\beta h (s_i + s_j)/4}. \quad (22)$$

Now the product goes over nearest neighbors, but since in a 2D square lattice each spin has four nearest neighbors, there has to appear a factor 4 for compensation. Now the partition function becomes

$$Z(\beta) = \sum_{\{s_i\}} \prod_{\langle i,j \rangle} e^{\beta J s_i s_j} \prod_i e^{\beta h s_i} = \sum_{\{s_i\}} \prod_{\langle i,j \rangle} e^{\beta J s_i s_j + \beta h (s_i + s_j)/4}. \quad (23)$$

In the Ising model the spins can only take the value $+1$ or -1 , so the exponent that appears in the product can take only four values, namely the value

$$Q_{s_i, s_j} = e^{\beta(J+h/2)} \quad \text{for } s_i = +1 \text{ and } s_j = +1, \quad (24)$$

$$Q_{s_i, s_j} = e^{-\beta J} \quad \text{for } s_i = -1 \text{ and } s_j = +1, \quad (25)$$

$$Q_{s_i, s_j} = e^{-\beta J} \quad \text{for } s_i = +1 \text{ and } s_j = -1 \quad \text{and} \quad (26)$$

$$Q_{s_i, s_j} = e^{\beta(J-h/2)} \quad \text{for } s_i = -1 \text{ and } s_j = -1. \quad (27)$$

I used the notation Q_{s_i, s_j} , because this information can be rearranged in a matrix Q , defined as

$$Q \equiv \begin{pmatrix} e^{\beta(J+h/2)} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta(J-h/2)} \end{pmatrix}. \quad (28)$$

With this definition, the final expression for the partition function becomes

$$Z(\beta) = \sum_{\{s_i\}} \prod_{\langle i,j \rangle} Q_{s_i, s_j}. \quad (29)$$

The partition function can now be displayed as a tensor network. A tensor network is a group of tensors which indices are connected in a certain way. In this notation, a tensor is displayed as a shape with lines coming out of it. These lines represent the indices of the tensor and can each have different dimensions. When two shapes are connected with a line, it means that the indices of these tensors are being contracted. The open lines of a tensor network are the indices of the remaining tensor if all the indices of the tensors have been contracted. In figure 1 is depicted how a scalar, vector and matrix are shown in this notation and how a multiplication of tensors looks like.

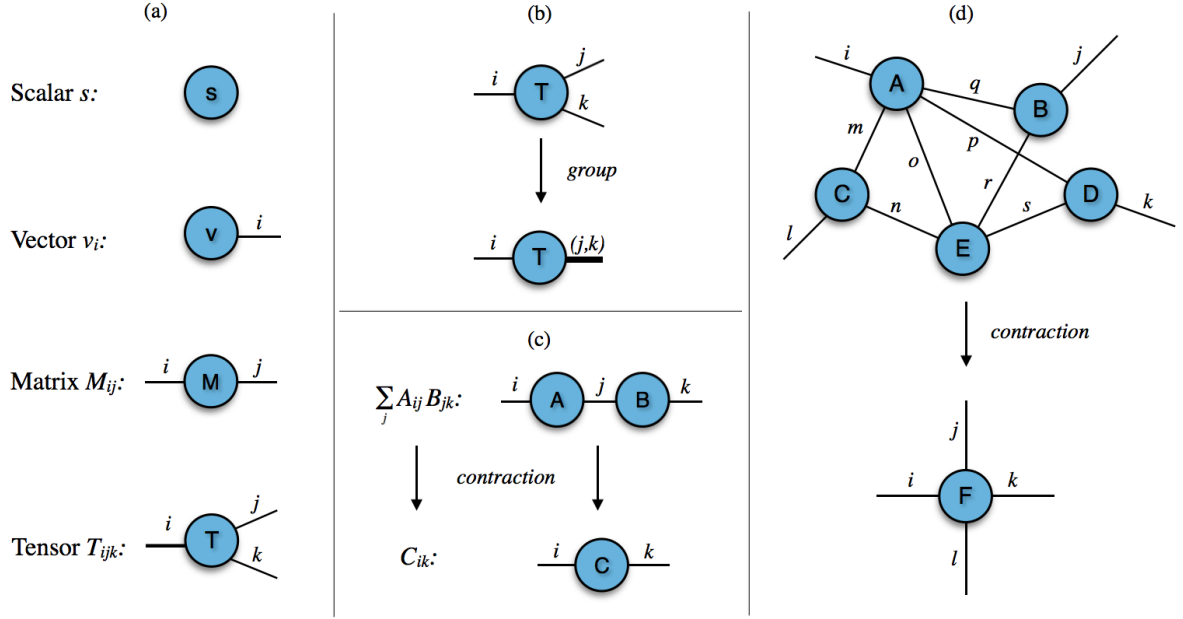


Figure 1: Diagrammatic notion of a tensor network. a) Diagram of a scalar, vector, matrix and a 3rd-order tensor. b) Diagram of a grouping of indices. c) Diagram of a multiplication of tensors/contraction of indices. d) Diagram of a contraction of a tensor network.

The partition function can now be displayed as a tensor network of tensors Q_{s_i, s_j} and δ_{ijkl} , where $\delta_{ijkl} = 1$ if $i = j = k = l$ and $\delta_{ijkl} = 0$ otherwise. In this network the tensor Q_{s_i, s_j} represents the interaction between spin i and spin j . The tensor δ_{ijkl} represents the spin on a certain lattice site. When introducing the tensor a_{ijkl} , given by

$$a_{ijkl} = \sum_s (\sqrt{Q})_{is} (\sqrt{Q})_{js} (\sqrt{Q})_{ks} (\sqrt{Q})_{ls}, \quad (30)$$

the tensor network will consist of a network of tensor a_{ijkl} . This is shown in figure 2.

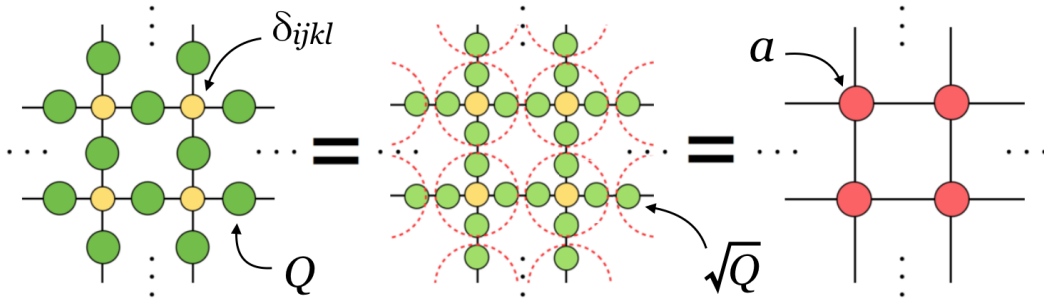


Figure 2: Diagrammatic notation of the partition function. The partition function is a tensor network of tensors Q , given by equation 28, and the Kronecker delta tensors δ_{ijkl} . The tensor Q can be written as $\sqrt{Q} \cdot \sqrt{Q}$. By contracting four $\sqrt{Q}$ tensors with δ_{ijkl} , the partition function only consists of tensors a .

The magnetization per site can also be expressed as a tensor network. Given the expression in equation 5 it is possible to follow a similar procedure as for the partition function in order to obtain

$$m = \frac{1}{Z} \sum_{\{s_i\}} s_r \prod_{\langle i,j \rangle} Q_{s_i, s_j}. \quad (31)$$

So m is a tensor network divided by another tensor network, the partition function. The tensor network in the numerator is very similar to the partition function. The only difference is that instead of a tensor a_{ijkl} there is a tensor b_{ijkl} at lattice site r which measures the spin value on that site. This tensor is defined by

$$b_{ijkl} = \sum_{s_r} s_r \left(\sqrt{Q} \right)_{is_r} \left(\sqrt{Q} \right)_{js_r} \left(\sqrt{Q} \right)_{ks_r} \left(\sqrt{Q} \right)_{ls_r}. \quad (32)$$

The diagrammatic notation for the magnetization per site is shown in figure 3.

$$m(\beta) = \frac{\sum_{\{c\}} s_r \exp(-\beta H(c))}{Z} = \frac{\text{Diagrammatic representation of } m(\beta)}{\text{Diagrammatic representation of } Z}$$

Figure 3: Diagrammatic notation of the magnetization per site. The magnetization per site is a division of two tensor networks. The network in the denominator consists only of tensors a , which are defined by equation 30. The network in the numerator is identical, but one tensor a is replaced by tensor b , which is defined by equation 32.

2.2 The Potts model

2.2.1 Definition of the Potts model

The Potts model is a generalization of the Ising model. Renfrey Potts described this model in 1951 after it was proposed by his advisor Cyril Domb [1, 2]. In this model the spins can be in one of the q different states $\{1, 2, \dots, q\}$, so it coincides the Ising model when $q = 2$. The Hamiltonian of the Potts model with a uniform interaction strength and no external magnetic field is given by

$$H(\{s_i\}) = -J \sum_{\langle i, j \rangle} \delta_{s_i s_j}. \quad (33)$$

The notation $\langle i, j \rangle$ again means that the sum goes over all nearest neighbor pairs of spins. $\delta_{s_i s_j}$ is the Kronecker delta function, which is 1 if s_i equals s_j and 0 otherwise. So when two neighboring spins are in the same state, they contribute $-J$ to the total energy. In contrast to the Ising model, they contribute nothing if they are in different spin states.

Just like the Ising model, the Potts model behaves as a ferromagnet when $J > 0$ and as a anti-ferromagnet when $J < 0$. In the ferromagnetic case the majority of spins will be in the same state when the temperature goes below the critical temperature T_c . If the majority is in state 1, this spontaneous magnetization can be expressed by

$$m = \frac{q\langle\delta_{1s}\rangle - 1}{q - 1}, \quad (34)$$

where $\langle\delta_{1s}\rangle$ is the spin density of state 1. The spontaneous magnetization is often called "the order parameter". The Potts model has an exact solution for the critical temperature [15], which is given by

$$T_c = \frac{J}{k_B \ln(1 + \sqrt{q})}. \quad (35)$$

Note that if you fill $q = 2$ in this formula you find that $T_c \approx 1.135 J/k_B$, which is exactly half of the critical temperature of the Ising model. This can be explained by the difference in the Hamiltonians of the Ising and Potts model. The energy spectrum of the Ising model is twice as big as that of the Potts model, which causes the temperature scale to stretch out with factor 2.

Baxter showed that for $q > 4$, the Potts model has a first-order phase transition [16], and the magnetization therefore shows a discontinuity near the critical temperature. For $q \leq 4$ the phase transition is of order higher than one and the magnetization therefore goes continuously to zero.

2.2.2 Tensor network representation of the 2D Potts model

Similarly as for the Ising model, the partition function of the Potts model can be expressed as a tensor network. In the absence of a magnetic field, the tensor $Q_{s_i, s_j} = \exp(\beta J \delta_{s_i s_j})$ can now take the values

$$Q_{s_i, s_j} = e^{\beta J} \quad \text{for } s_i = s_j \quad \text{and} \quad (36)$$

$$Q_{s_i, s_j} = 1 \quad \text{for } s_i \neq s_j. \quad (37)$$

This information can be rearranged in the matrix Q defined by equation 38. All the off-diagonal elements of this matrix are ones and the diagonal elements are $\exp(\beta J)$.

$$Q \equiv \begin{pmatrix} e^{\beta J} & 1 & \dots & 1 \\ 1 & e^{\beta J} & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \dots & e^{\beta J} \end{pmatrix} \quad (38)$$

The partition function of the Potts model can now be calculated by using this new definition for the tensor Q in equation 4. In order to use equation 34 for calculating the magnetization, an expression for $\langle \delta_{1s} \rangle$ is needed. A tensor which measures the spin density of state 1 at site r is given by

$$b_{ijkl} = \sum_{s_r} \delta_{1s_r} \left(\sqrt{Q} \right)_{i s_r} \left(\sqrt{Q} \right)_{j s_r} \left(\sqrt{Q} \right)_{k s_r} \left(\sqrt{Q} \right)_{l s_r}, \quad (39)$$

where $\delta_{1s_r} = 1$ if the spin state at site r is 1 and $\delta_{1s_r} = 0$ otherwise. With this new definition of b_{ijkl} it is possible to calculate $\langle \delta_{1s} \rangle$ (and thus m) using the same calculation for the magnetization of the Ising model shown in figure 3.

3 Method

3.1 Simulation of the partition function

I made an algorithm in Python that can calculate several thermodynamic quantities of the 2D Ising and Potts model in zero magnetic field ($h = 0$). In these calculations I set the interaction energy and Boltzmann's constant equal to 1 ($J = k_B = 1$). In order to find the partition function as function of temperature, I used the CTMRG method [9]. Here I explain how this method works step by step.

As you can see from figure 2, the partition function of the Ising model is a tensor network that consist of infinitely many tensors a , given by equation 30. The first approximation in the CTMRG method is that the environment of a particular lattice site can be represented by corner tensors C and edge tensors T , see figure 4. This approximation is exact if the bond dimensions of these tensors are infinite. Tensor C is a corner transfer matrix that takes into account one quadrant of the infinite system and tensor T takes into account an infinite half-row/column of the system. Because of the rotational and translational symmetry in the lattice, all corners can be represented by the same tensor C and all edges by the same tensor T .

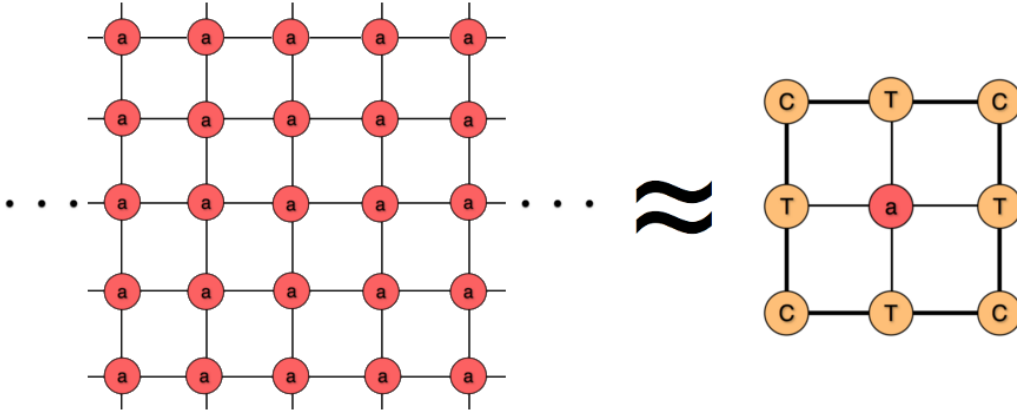


Figure 4: Diagram of the partition function. The tensor network on the left can be approximated by the tensor network on the right by using corner transfer matrices C and T . This approximation becomes exact if the bond dimensions of tensor C and T are infinite.

The matrices C and T are obtained in an iterative way. The program starts with the tensor network on the right of figure 4, where C and T are the tensors

$$C_{ij} = \sum_s \left(\sqrt{Q} \right)_{is} \left(\sqrt{Q} \right)_{js} = Q_{ij} \text{ and} \quad (40)$$

$$T_{ijk} = \sum_s \left(\sqrt{Q} \right)_{is} \left(\sqrt{Q} \right)_{js} \left(\sqrt{Q} \right)_{ks} \text{ respectively.} \quad (41)$$

For the Ising model, Q is defined by equation 28 and for the Potts model it is defined by equation 38. This tensor network represents the partition function of a 3×3 square lattice. The next step is to grow the system by repeating the following procedure. In each iteration, a row and column is added to tensor a consisting only of tensors a . The boundary consists again of the corner and edge tensors C and T . In other words, the system grows from a 3×3 lattice to a 5×5 lattice. Then the tensor network is cut into nine pieces. These steps are shown in figure 5.

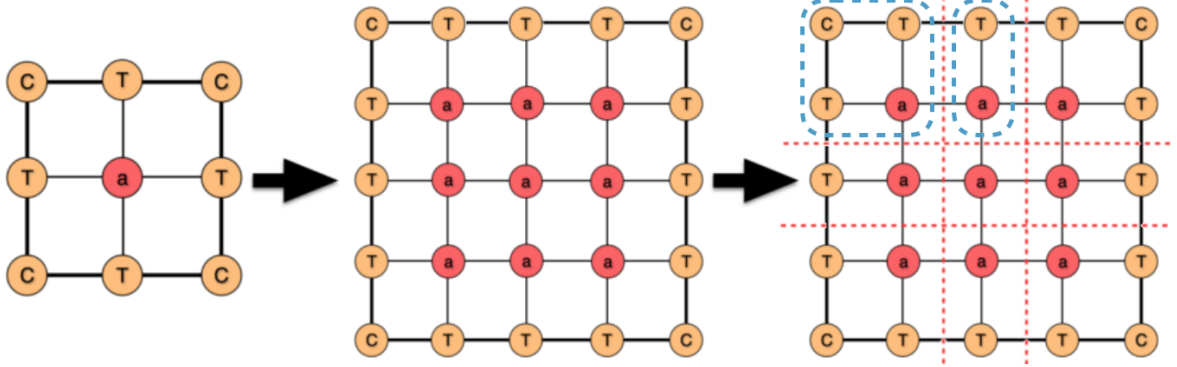


Figure 5: Growth of the lattice per iteration. The lattice grows from $N \times N$ to $(N+2) \times (N+2)$, by adding eight new tensors a to tensor a , with edge tensors T at the boundary. Then the network is cut into nine pieces along the red dotted lines. Because of the rotational symmetry in the lattice, only two of the eight boundary pieces are different. These are indicated with the blue dotted regions.

In principle it is possible to contract both boundary tensors and use them as new environment tensors C and T for the next iteration. But the problem is that in each iteration the dimensions of the tensors grow with factor q , where q is again the number of states the spins can be in. That means that the computation time grows exponentially with the number of iterations. This is why a renormalization is needed.

In my program I use the boundary tensors in the following way. From the contracted corner tensor M a singular value decomposition is made, which is shown in figure 6. This decomposition factorizes the tensor into the product $U \cdot s \cdot V^T$, where U and V^T are unitary matrices. s is a diagonal matrix with on its diagonal the singular values of the original matrix in descending order. The idea behind the renormalization is that the exact tensor M can be represented by another tensor $\tilde{M}$ that is of rank χ , where χ is smaller than the original rank. Performing a singular value decomposition of M and keeping only the largest χ singular values gives the best lower rank approximation. In other words, it minimizes the Frobenius norm $\|M - \tilde{M}\|$. A similar idea is used in the DMRG method [8] to efficiently represent quantum many-body states in an exponentially large Hilbert space.

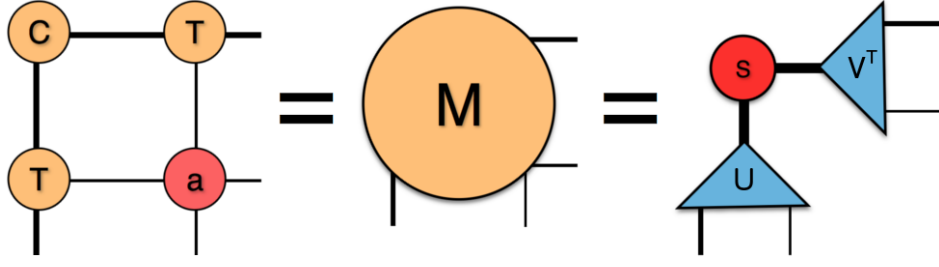


Figure 6: The boundary tensor is contracted to tensor M . Of M a singular value decomposition is made: $M = U \cdot s \cdot V^T$.

The dimensions of the boundary tensors are reduced by multiplying the tensors from both sides by $\tilde{U}^\dagger$, as you can see in figure 7. This is possible because U and V^T are equivalent up to sign factors. The resulting tensors $\tilde{C}$ and $\tilde{T}$ have bond dimensions χ and are used for the next iteration. This process continues until the tensors have converged. The way how the program determines when the tensors have converged is by evaluating the sum of the singular value spectrum after each step and comparing it with the sum of the previous step. When this difference is below a given tolerance, the tensors C and T are used for calculating thermodynamic quantities.

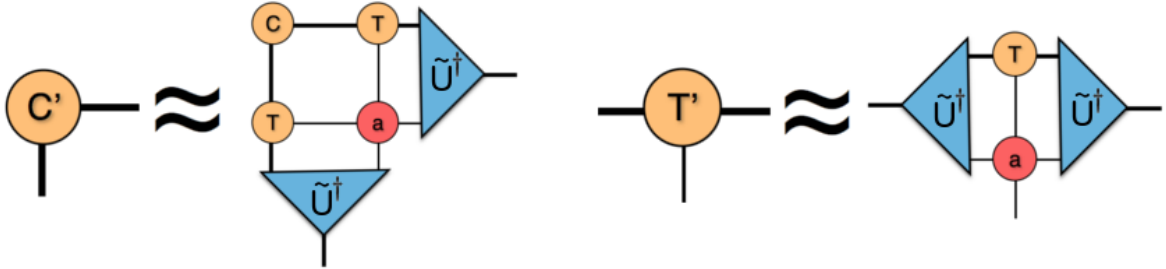


Figure 7: Renormalization of the boundary tensors. The new environment tensors C' and T' are obtained by multiplying the boundary tensors by the reduced tensor $\tilde{U}^\dagger$.

3.2 Simulation of the magnetization per site

The magnetization per site/order parameter of the Ising and Potts model can be calculated when the converged environment tensors C and T are obtained by the CTMRG method. The diagram of the magnetization in figure 3 can now be displayed by figure 8.

$$m(\beta) = \frac{\sum_{\{c\}} s_r \exp(-\beta H(c))}{Z} \approx \frac{\text{Diagram 1}}{\text{Diagram 2}}$$

Figure 8: Approximated diagrammatic notation of the magnetization. The environment of tensors a and b are approximated by environment tensors C and T .

The tensors a and b that I use for the Ising model are defined by equation 28, 30 and 32. For the Potts model they are defined by equation 30, 38 and 39. In order to use tensor b in the Potts model, the system has to fall in the spin 1 state when $T < T_c$. This is done by starting the simulation with the environment tensors

$$C_{ij} = \sum_s \delta_{1s} \left(\sqrt{Q} \right)_{is} \left(\sqrt{Q} \right)_{js} \quad \text{and} \quad (42)$$

$$T_{ijk} = \sum_s \delta_{1s} \left(\sqrt{Q} \right)_{is} \left(\sqrt{Q} \right)_{js} \left(\sqrt{Q} \right)_{ks}. \quad (43)$$

With these tensors the boundary of the system is in the spin 1 state, which in the end does not matter because the boundary effects disappear when the number of iterations increases.

For both the Ising and the Potts model I calculate the magnetization as function of temperature for different bond dimension χ and compare the results with the exact solution, given by equation 19. I also study how the tolerance for the converging environment tensors affects the shape of the acquired graphs. Then I determine the critical temperature T'_c for different χ , which I define as the temperature at which the slope of the magnetization is steepest. From these data I plot T'_c against χ and do a fit in order to determine the real critical temperature T_c .

For the Ising model I calculate the critical exponent of the magnetization (β). I do this by calculating the magnetization in a small temperature interval around T'_c . In order to make this result accurate, I choose a high bond dimension ($\chi = 60$), a small tolerance of 10^{-8} and temperature steps of $\sim 10^{-5}$. Then I do a fit of the form of equation 13 in order to determine β .

3.3 Simulation of the free energy per site

The free energy per site is given by equation 6 and can be calculated after simulation of the partition function per site κ , which is given by

$$\kappa = \lim_{N \rightarrow \infty} Z^{1/N}. \quad (44)$$

Figure 9 shows how the partition function per site is calculated [17]. When the logarithm of κ for a given temperature is obtained, I multiply this value by the temperature and take the negative in order to get the free energy per site. I calculate the free energy as function of temperature for different χ and compare the results with Onsager's solution, given by equation 20.

$$\kappa = \lim_{N \rightarrow \infty} Z^{1/N} \approx \frac{\text{Diagram 1} \times \text{Diagram 2}}{\left(\text{Diagram 3} \right)^2}$$

Figure 9: Approximated diagrammatic notation of the partition function per site κ . The partition function per site can be represented by a division of tensor networks, consisting of tensor a and approximated environment tensors C and T .

3.4 Simulation of the specific heat per site

For the Ising model, I calculate the specific heat per site and its critical exponent (α). The specific heat is related to the partition function by equation 7. In order to calculate the specific heat, I calculate $\ln(\kappa)$ as function of temperature, where κ is again the partition function per site. For this calculation I take $\chi = 60$ and tolerance = 10^{-8} . Of the acquired graph I take the second derivative with respect to the temperature by using that

$$\frac{\partial^2}{\partial T^2} (\ln(\kappa(T))) = \lim_{\Delta T \rightarrow 0} \frac{\ln(\kappa(T - \Delta T)) - 2\ln(\kappa(T)) + \ln(\kappa(T + \Delta T))}{\Delta T^2}, \quad (45)$$

where I approximate the limit by using the value $\Delta T \sim 10^{-5}$. After taking the second derivative I only have to multiply by the temperature and take the negative in order to determine the specific heat. Then I show that the specific heat diverges logarithmically by fitting the graph with a function of the form of equation 12.

3.5 Simulation of the magnetic susceptibility per site

The last thermodynamic quantity that I calculate for the Ising model is the magnetic susceptibility per site. This quantity can be calculated after simulation of the magnetization. The relation between χ_m and m is given by equation 8. In order to take the derivative of m with respect to h I apply a small magnetic field; I use the value $h' = 10^{-8}$ in equation 28. Then I compute χ_m by using that

$$\chi_m(T) = \left. \frac{\partial m}{\partial h} \right|_{h=0} \approx \frac{m(T, h = h') - m(T, h = 0)}{h' - 0}. \quad (46)$$

Finally, I calculate the critical exponent of the magnetic susceptibility (γ). I do this by computing χ_m as function of temperature in a small interval around T_c . In order to make the result more accurate, I take $\chi = 60$, tolerance = 10^{-7} and temperature steps $\sim 10^{-5}$. For the graph I use a fit of the form of equation 14 in order to determine γ .

4 Results

4.1 Magnetization per site

For the Ising model I have calculated the magnetization per site as function of temperature for different bond dimensions χ , which is shown in figure 10. From this figure you can see that the simulated data points coincide the exact points more and more when χ increases. This is because the correlation length is cut off by the finite χ . This effect is more important for smaller χ , since in that case more information about the system is deleted in the simulation.

In figure 11 you can see the magnetization per site as function of temperature for $\chi = 11$ for the 3,4,6- and 7-state Potts model. This figure shows that for all models the magnetization goes to zero near the exact critical temperature. From the discontinuity below the T'_c you can also see that the phase transition of the 6- and 7-state Potts model is of first order.

I have also plotted the magnetization of the Ising model for $\chi = 6$ for different tolerances, which is shown in figure 12. From this plot you can see that the 'tail' of the curve becomes smaller when the tolerance is decreased. This can be explained by the fact that a smaller tolerance corresponds to a larger system size, because the environment tensors converge longer. The cut-off at T_c becomes sharper when the system size goes to infinity.

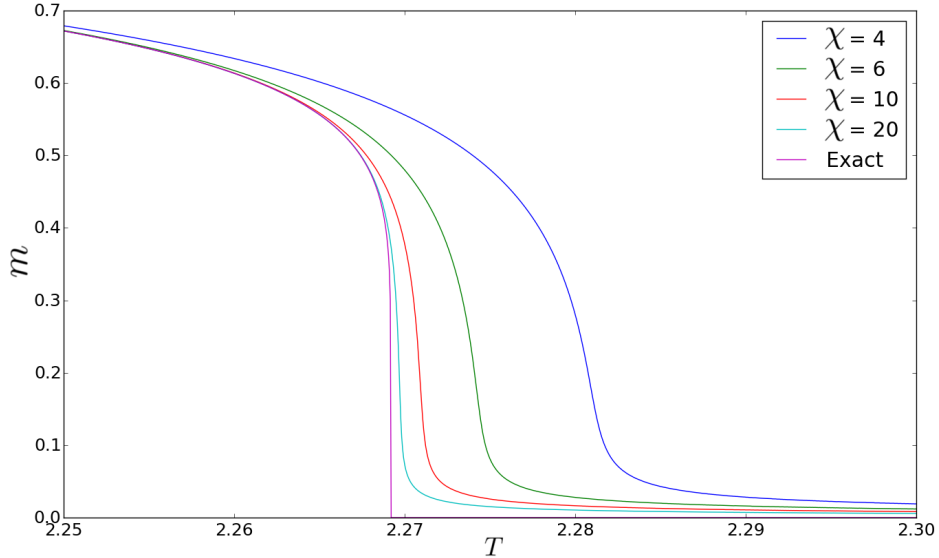


Figure 10: Simulation of the magnetization per site for the Ising model. The graph shows the magnetization per site as function of temperature for different bond dimensions χ . The exact curve is also indicated.

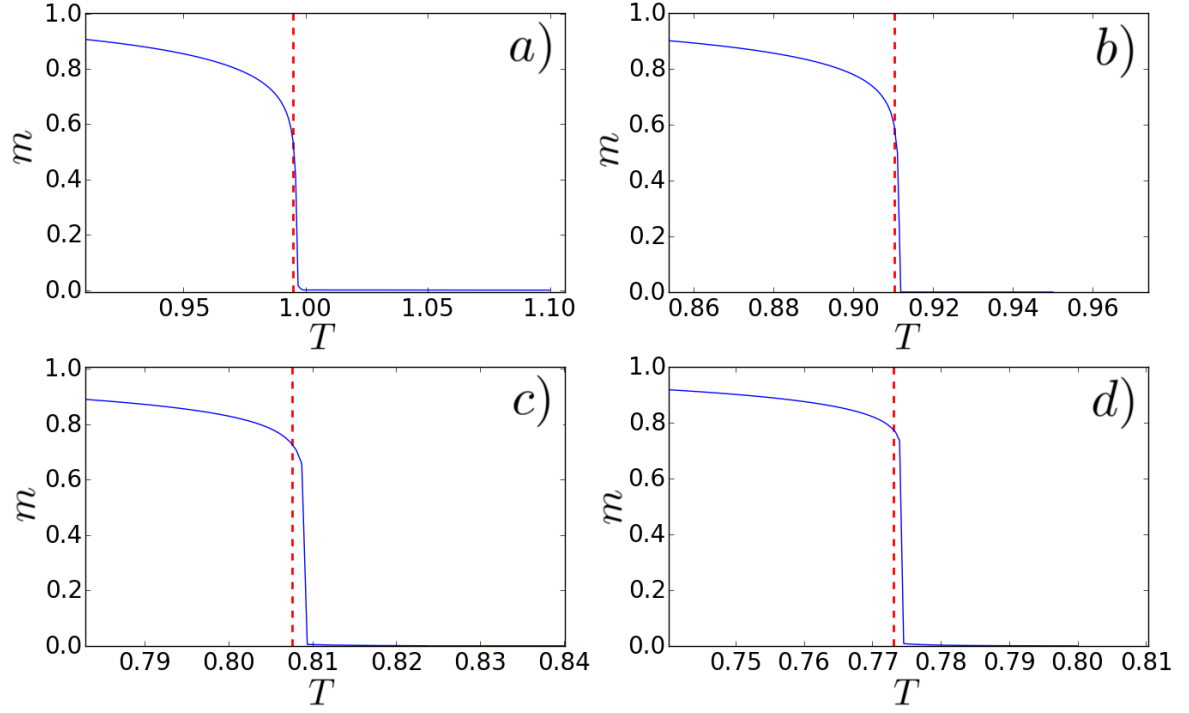


Figure 11: Simulation of the magnetization per site for the q -state Potts model. The magnetization is plotted as function of temperature for $\chi = 11$ for the models: $q = 3$ (a), $q = 4$ (b), $q = 6$ (c) and $q = 7$ (d). The exact T_c is indicated in each graph with the red dotted line.

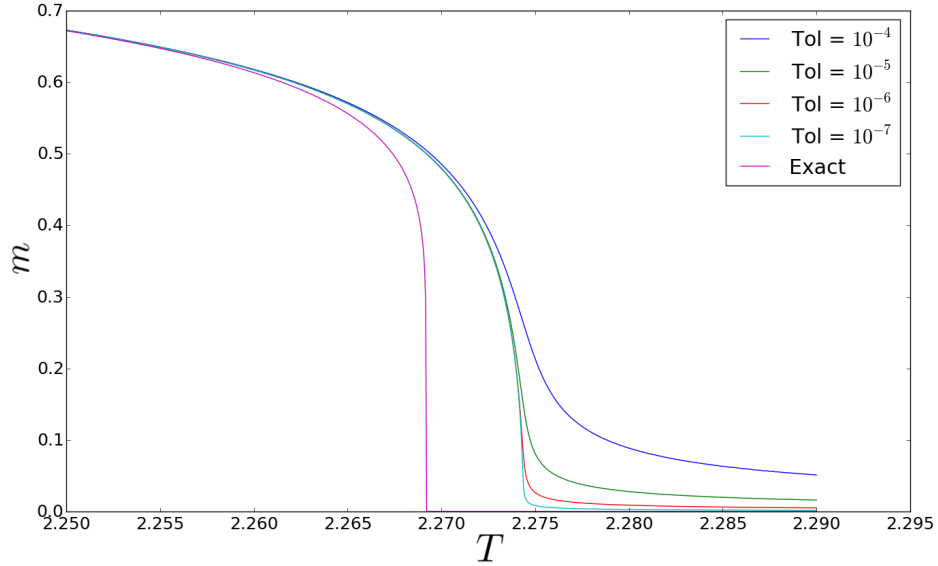


Figure 12: Simulation of the magnetization per site for the Ising model. The graph shows the magnetization per site as function of temperature for bond dimension $\chi = 6$ for different tolerances. The exact solution for the magnetization is also included.

The critical temperature T'_c as function of χ is shown in figure 13 for the Ising model and in figure 14 for the 3-state Potts model. These data suggests that $|T'_c - T_c|$ goes with a power law. Therefore I fitted the data with the function $T'_c(\chi) = T_c + a \cdot \chi^b$. The T'_c determined for the highest bond dimension I used in the simulations is shown in table 1, along with the exact solution and the T_c determined from the fit. From this table you can see that T'_c already is really close to the exact value for $\chi = 60$. The T_c determined from the fit is even closer.

	T'_c for $\chi = 60$	T_c from fit	Exact T_c
Ising model	2.26925(1)	2.26920(1)	$2/\ln(1 + \sqrt{2}) \approx 2.26919$
3-state Potts model	0.99503(1)	0.99499(3)	$1/\ln(1 + \sqrt{3}) \approx 0.99497$

Table 1: Values of the critical temperature for the Ising and 3-state Potts model. Shown for both models are the critical temperature T'_c determined for $\chi = 60$, the critical temperature T_c determined by the fit and the exact value.

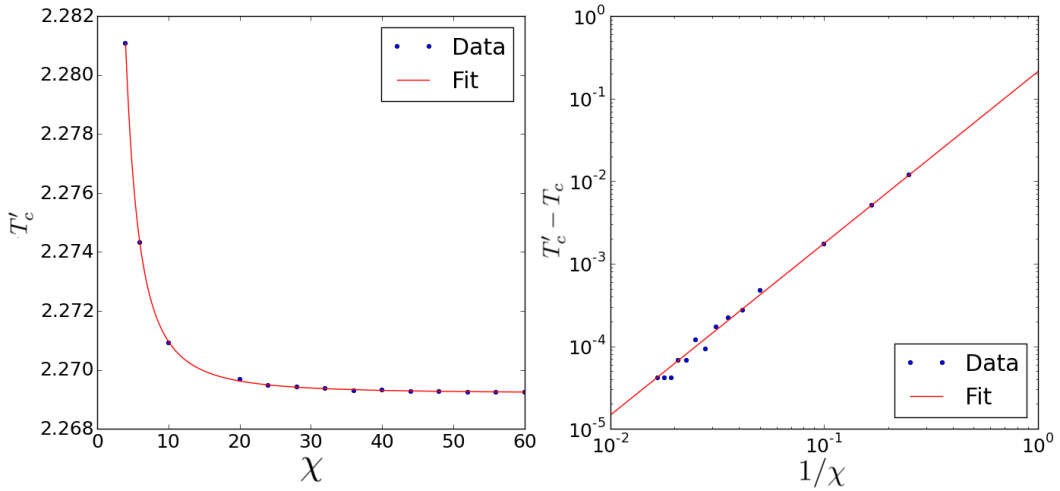


Figure 13: Determination of the critical temperature of the Ising model. Left: the critical temperature is plotted against the bond dimension, along with the fitted curve. Right: the difference between the critical temperature and the one determined by the fit is plotted against one over the bond dimension.

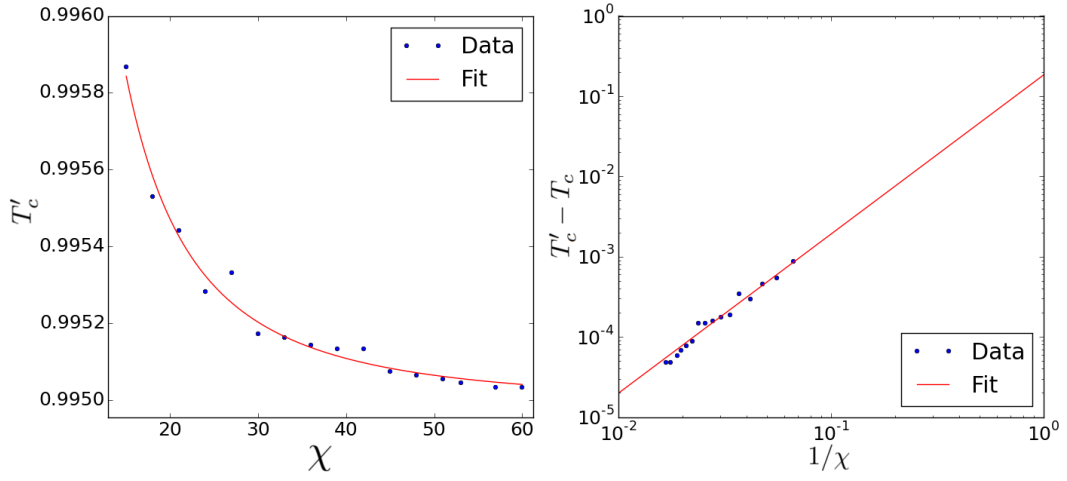


Figure 14: Determination of the critical temperature of the 3-state Potts model. Left: the critical temperature is plotted as function of the bond dimension, along with the fitted curve. Right: the difference between the critical temperature and the one determined by the fit is plotted as function of one over the bond dimension.

The critical exponent β is determined by fitting the data points in figure 15. From the fit, I found that $\beta = 0.1264(4)$.

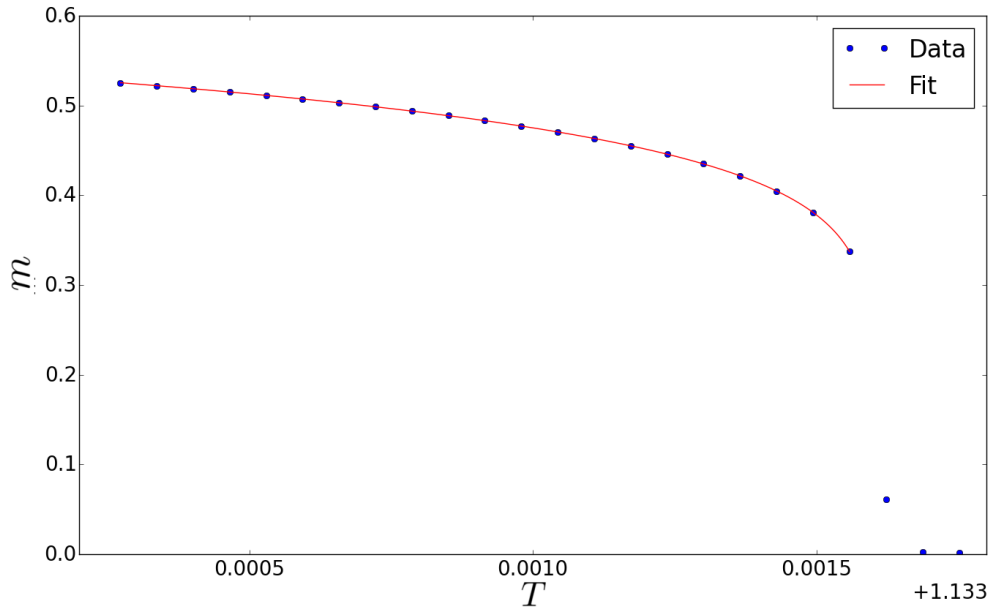


Figure 15: Determination of the critical exponent of the magnetization of the 2D Ising model. The simulated data points of the magnetization for bond dimension $\chi = 60$ is plotted as function of temperature. The fitted curve is also indicated.

4.2 Free energy per site

The left part of figure 16 displays the free energy per site of the Ising model as function of temperature for different χ . Onsager's solution, given by equation 20, is also indicated. The errors of these results are shown in the right part of figure 20. A remarkable thing is that the results are very accurate, even for small χ .

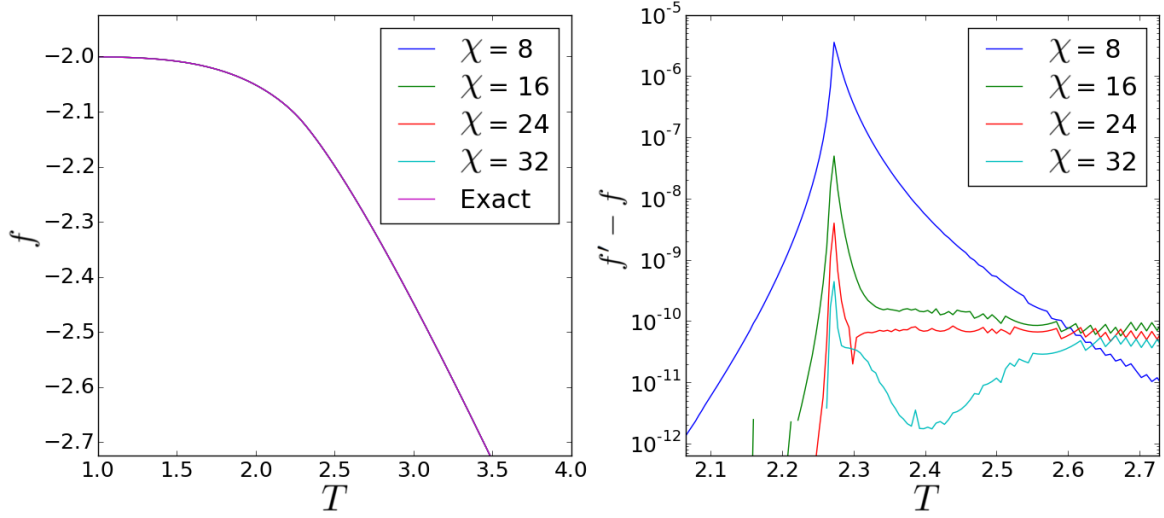


Figure 16: Simulation of the free energy of the Ising model. Left: the free energy per site is plotted as function of temperature for different bond dimensions χ . Onsager's exact solution [13] is also indicated. Right: the difference between the simulated free energy and Onsager's solution for the free energy is plotted as function of temperature.

4.3 Specific heat per site

The specific heat per site for the Ising model is shown in figure 17 for $\chi = 60$, along with a fitted curve and Onsager's exact solution. A remarkable thing is that the specific heat goes with a logarithm around T_c just like the exact solution, but does not diverge at T_c . The reason for this behavior is that the correlation length can not diverge at T_c , because the system is not really infinite and the bond dimension is truncated to a finite χ . The logarithmic behavior establishes the fact that for the critical exponent of the specific heat holds that $\alpha = 0$.

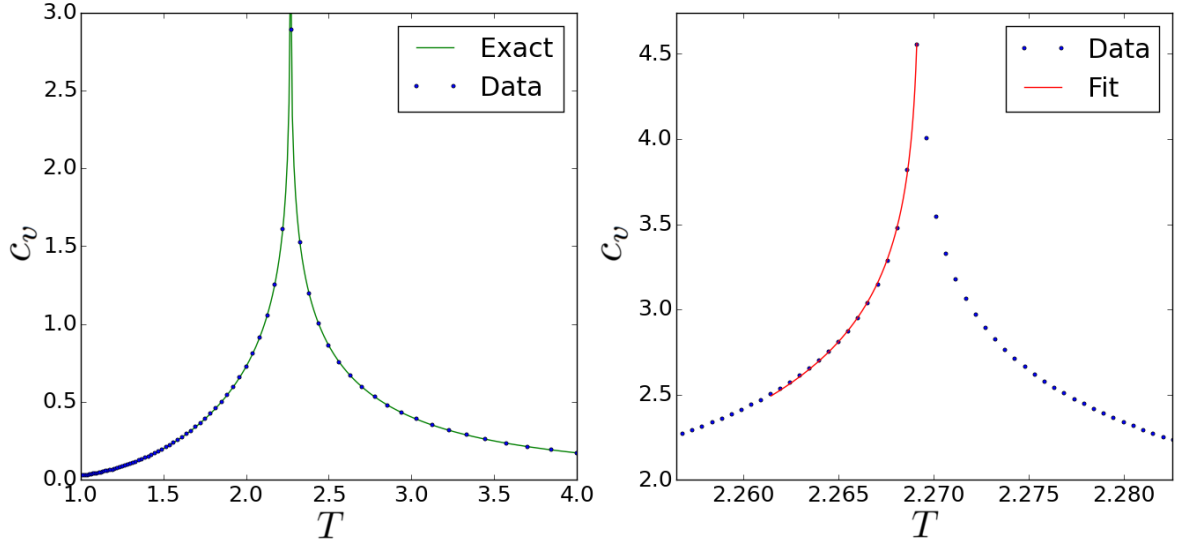


Figure 17: Simulation of the specific heat of the Ising model. Left: the specific heat per site is plotted as function of temperature for $\chi = 60$. Onsager's exact solution is also indicated. Right: a zoomed picture of the left picture around the critical temperature with a fitted curve.

4.4 Magnetic susceptibility per site

The magnetic susceptibility per site of the Ising model is shown in figure 18 for $\chi = 60$. After fitting the data points, I have found that the critical exponent of the magnetic susceptibility has the value $\gamma = 1.57(9)$. Just like the specific heat, the magnetic susceptibility does not go to infinity at the critical temperature. This could again be explained by the fact that the correlation length does not diverge in the simulations.

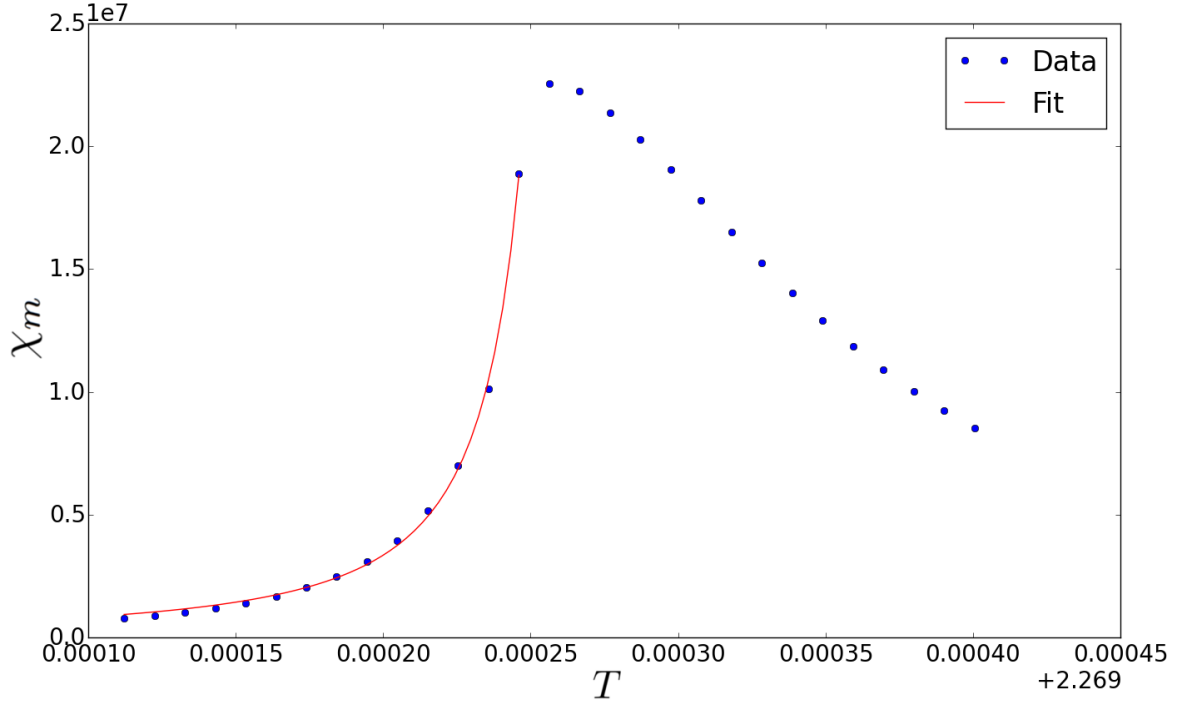


Figure 18: Simulation of the magnetic susceptibility of the Ising model. The magnetic susceptibility per site is plotted as function of temperature for $\chi = 60$. Also indicated are the fitted curve around the critical temperature.

5 Conclusion

Using the CTMRG method, I simulated the partition function of the 2D square lattice Ising model. With the partition function of the Ising model I calculated the magnetization, free energy, specific heat and the magnetic susceptibility per site as function of temperature and determined the critical temperature. In order to test the method, I compared the results with the exact solutions. Then I studied the magnetization of the Potts model.

From the simulation of the magnetization, I found for the critical temperature of the Ising model that $T_c = 2.26920(1)$. This value is really close to the exact value of $T_c \approx 2.26919$, which shows that the method is very accurate. For the critical exponent of the magnetization I found the value $\beta = 0.1264(4)$, which is close to the exact value of $\beta = 1/8$. In the same way as I did for the Ising model I determined the critical temperature for the 3-state Potts model and found that $T_c = 0.99499(3)$. This value agrees with the exact value of $T_c \approx 0.99497$.

After comparing the free energy per site with the exact solution, I found that they were very similar. The free energy approached the exact solution within an error of 10^{-5} for $\chi = 8$ and this error decreased to 10^{-9} when the bond dimensions increased to $\chi = 32$.

Just like the exact solution, the specific heat per site behaved logarithmically around T_c . It did however not diverge at T_c . This can be explained by the facts that the size of the system is not really infinite in the simulations and the bond dimensions are finite. Therefore, the correlation length can not go to infinity. From the logarithmic behavior around T_c I conclude that the critical exponent of the specific heat is $\alpha = 0$.

Finally, I simulated the magnetic susceptibility per site and determined its critical exponent. I found the value $\gamma = 1.57(9)$, which is similar to the exact value of $\gamma = 7/4$. Just like the specific heat, the magnetic susceptibility does not really diverge at the critical temperature, but this can again be explained by the facts that the system is finite and not all the information about the system is kept because of the finite χ .

I can conclude that the CTMRG method is a good method in order to calculate properties of the Ising model, because the results of the simulations matched the exact solutions really well. The results can be even more accurate by taking higher bond dimensions, smaller temperature steps and letting the transfer matrices converge longer, but this of course takes longer computation time. A natural next step from here would be to perform a systematic finite-size and finite-bond dimension scaling analysis in order to get more accurate results. After studying the magnetization of the Potts model, another step can be to calculate other thermodynamic quantities of the Potts model. Another interesting thing is to generalize the algorithm such that it works also for the 3D Potts model.

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